

Data Collection, Cleaning and Visualization

Learning Objectives

- Understand data collection methods
- Learn sources of data
- Understand data quality issues
- Learn basic visualization techniques
- Interpret visual data for decision making

What is Data Collection?

- Process of gathering and measuring information from various sources.
- Example: Collecting daily temperature data from weather stations.

Types of Data Collection

1. Primary Data

- Collected directly by researcher
- Surveys, experiments, observations

2. Secondary Data

- Already collected by others
- Government reports, databases

Primary Data Example – Survey

- Example: Student Satisfaction Survey
- Questions:
 - Rate teaching quality (1–5)
 - Rate infrastructure (1–5)
 - Rate library facilities (1–5)
- Collected from 100 students.

Primary Data Example – Sensors

- Example: Smart Agriculture

Sensors collect:

- Soil moisture
 - Temperature
 - Humidity
-
- Used to optimize irrigation.

IOT devices in oil and chemical industries

- Pressure and Temperature Sensors: Monitor pressure and temperature levels in pipelines, storage tanks, and processing equipment (**Honeywell ST3000**)
- Flow Meters: measure the rate of fluid movement in pipelines, especially important in transporting oil, gas, and chemicals. (**Siemens SITRANS FS230**)
- Gas Leak Detectors: Identify the presence of hazardous gases, like methane, sulfur dioxide, or hydrogen sulfide, in real-time. (**MSA Altair 4X**)
- Environmental and Emission Monitors: Monitor emissions from factories, refineries, and other equipment, tracking pollutants to ensure regulatory compliance. (Aeroqual Series 500, Thermo Fisher 42i NO-NO₂-NO_x Analyzer)
- Vibration Sensors (SKF CMSS 2200), Level Sensors (**VEGA VEGAPULS 64**), Wearable IoT Devices for Safety, GPS Trackers and Fleet Management Devices etc.

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Secondary Data Example

- Example: Using Government Open Data
- Dataset: City population data
- Year Population
- 2020 1,200,000
- 2021 1,260,000
- 2022 1,330,000

Data Collection Workflow

- Define problem
- Identify data sources
- Collect data
- Clean and validate data
- Store data
- Analyze and visualize

Data Quality Issues

- Common problems:

Missing values

Duplicate records

Incorrect measurements

Inconsistent formats

Example: Age recorded as 200 years.

Real World Case Study

- E-commerce company collects:

Customer purchases

Website clicks

Product ratings

- Visualizations used:

Sales trend graphs

Product popularity charts

Customer segmentation plots

Data Cleaning Techniques

- Handling Missing Values
- Removing Duplicate Data
- Handling Outliers (Z-score method, IQR (Interquartile Range), Visualization (box plot))
- Standardization / Normalization (Min-Max Normalization)
- Correcting Inconsistent Data
- Data Transformation
- Noise Reduction (Clustering)
- Data Integration Cleaning
- Data Validation (Ensure data follows rules.)
- Feature Encoding (Convert Categorical data into numerical)

❖ Handling outliers using Z-score method

Observation	Sales
1	10
2	12
3	11
4	13
5	12
6	14
7	50

❖ Apply IQR for outlier handling

12,15,18,19,20,21,22,23,24,80

Lower Bound

$$\text{Lower Bound} = Q_1 - 1.5 \times IQR$$

Upper Bound

$$\text{Upper Bound} = Q_3 + 1.5 \times IQR$$

where,
 $IQR = Q_3 - Q_1$

Any value:

- less than Lower Bound, or
 - greater than Upper Bound
- is considered an outlier.

Data Cleaning Workflow

Raw Data



Data Inspection



Handle Missing Values



Remove Duplicates



Handle Outliers



Normalize / Transform Data



Clean Dataset

Example:

Student ID	Study Hours	Exam Score
S1	2	45
S2	4	60
S3	?	55
S4	6	70
S5	8	85
S6	4	60
S7	10	200

Introduction to Data Visualization

- Data visualization converts data into graphical form.

- Purpose:

Understand patterns

Detect trends

Communicate insights

Common Visualization Types

- Bar Chart – compare categories
- Line Chart – show trends over time
- Pie Chart – show proportions
- Scatter Plot – show relationships

Bar Chart Example

- Student Enrollments
- Course Students
- AI 120
- Data Sci 90
- ML 110
- Cyber Sec 70
- Bar chart helps compare enrollments.

Line Chart Example

- Monthly Sales Data
- Jan 10k
- Feb 15k
- Mar 18k
- Apr 25k
- Line chart shows increasing trend.

Scatter Plot Example

- Study Hours vs Exam Score
- Hours Score
- 2 50
- 4 65
- 6 75
- 8 90
- Scatter plot shows positive relationship.

Summary

- Data collection is the first step in analytics
- Data can be primary or secondary
- Data must be cleaned before analysis
- Visualization helps understand patterns
- Used widely in business, science, and research